

DBMS WEEKLY TASK

Meesho Jewellery Marketplace Database

Product Review and Rating Management System

1. Objectives

1. Create Review and Rating tables.
2. Store customer feedback and ratings.
3. Retrieve product review details.
4. Calculate average product ratings using aggregate functions.
5. Identify highly rated products.

2. REVIEW Table

The **REVIEW** table stores customer feedback about products.

Attribute	Data Type	Key / Meaning
REVIEW_ID	NUMBER(5)	Primary Key
CUSTOMER_ID	NUMBER(5)	Foreign Key
PRODUCT_ID	NUMBER(5)	Foreign Key
REVIEW_TEXT	VARCHAR2(500)	Customer Review
REVIEW_DATE	DATE	Date of Review

3. RATING Table

The **RATING** table stores product ratings separately from reviews.

Attribute	Data Type	Key / Meaning
RATING_ID	NUMBER(5)	Primary Key
CUSTOMER_ID	NUMBER(5)	Foreign Key
PRODUCT_ID	NUMBER(5)	Foreign Key
RATING	NUMBER(1)	Rating (1–5)
RATING_DATE	DATE	Date of Rating

4. Create Table

4.1 Review Table

```
SQL> CREATE TABLE REVIEW (  
2     REVIEW_ID NUMBER(5) PRIMARY KEY,  
3     CUSTOMER_ID NUMBER(5),  
4     PRODUCT_ID NUMBER(5),  
5     REVIEW_TEXT VARCHAR2(500),  
6     REVIEW_DATE DATE,  
7     CONSTRAINT FK_REVIEW_CUSTOMER  
8         FOREIGN KEY (CUSTOMER_ID)  
9         REFERENCES CUSTOMER(CUSTOMER_ID),  
10    CONSTRAINT FK_REVIEW_PRODUCT  
11        FOREIGN KEY (PRODUCT_ID)  
12        REFERENCES PRODUCT(PRODUCT_ID)  
13 );
```

Table created.

4.2 Rating Table

```
SQL> CREATE TABLE RATING (  
2     RATING_ID NUMBER(5) PRIMARY KEY,  
3     CUSTOMER_ID NUMBER(5),  
4     PRODUCT_ID NUMBER(5),  
5     RATING NUMBER(1) NOT NULL,  
6     RATING_DATE DATE NOT NULL,  
7     CONSTRAINT FK_RATING_CUSTOMER  
8         FOREIGN KEY (CUSTOMER_ID)  
9         REFERENCES CUSTOMER(CUSTOMER_ID),  
10    CONSTRAINT FK_RATING_PRODUCT  
11        FOREIGN KEY (PRODUCT_ID)  
12        REFERENCES PRODUCT(PRODUCT_ID),  
13    CONSTRAINT CHK_RATING  
14        CHECK (RATING BETWEEN 1 AND 5)  
15 );
```

Table created.

5. Retrieve Review Details

5.1 Retrieve Review details from customer

CUSTOMER_ID	REVIEW_ID	REVIEW_TEXT	REVIEW_DATE
1	1	Good quality necklace	01-OCT-26
2	2	Beautiful earrings	02-OCT-26
3	3	Nice bracelet	03-OCT-26
4	4	Good quality anklet	04-OCT-26
5	5	Excellent ring	05-OCT-26
1	6	Good quality necklace	01-OCT-26
2	7	Beautiful earrings	02-OCT-26
3	8	Nice bracelet	03-OCT-26
4	9	Good quality anklet	04-OCT-26
5	10	Excellent ring	05-OCT-26

5.2 Retrieve Rating details from product

PRODUCT_ID	PRODUCT_NAME	RATING_ID	RATING	RATING_DA
1	Gold Plated Necklace	1	5	01-OCT-26
2	Diamond Earrings	2	4	02-OCT-26
3	Pearl Bracelet	3	5	03-OCT-26
4	Silver Anklet	4	3	04-OCT-26
5	Stone Ring	5	4	05-OCT-26

6 . Calculate Average Product Rating

6.1 Average Rating

PRODUCT_ID	PRODUCT_NAME	AVERAGE_RATING
1	Gold Plated Necklace	5
3	Pearl Bracelet	5
5	Stone Ring	4
2	Diamond Earrings	4
4	Silver Anklet	3

6.2 Selected Average Rating

PRODUCT_ID	AVERAGE_RATING
3	5
1	5
5	4
2	4

7. Identify Highly Rated Products

PRODUCT_ID	PRODUCT_NAME	AVERAGE_RATING
3	Pearl Bracelet	5
1	Gold Plated Necklace	5
5	Stone Ring	4
2	Diamond Earrings	4

8. Conclusion

The Product Review and Rating Management System successfully manages customer reviews and product ratings in the Meesho Jewellery Marketplace Database. Separate REVIEW and RATING tables are used to store feedback and ratings. JOIN operations are used to retrieve the required details, while the AVG() aggregate function helps calculate average ratings and identify highly rated products.